08. Even More Transformations of Random Variables

January 23, 2026 8:53 AM

Last Time: Box-Muller Transform

if $U \sim Unif[0,1]$, $V \sim Exp(1)$ and $U \perp\!\!\!\perp V$

Then for $X = \sqrt{2V}\cos(2\pi U)$

$Y = \sqrt{2V}\sin(2\pi U)$

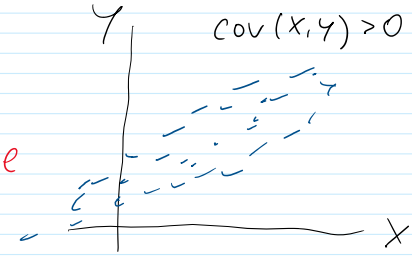
$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)$$ ← Bivariate Normal

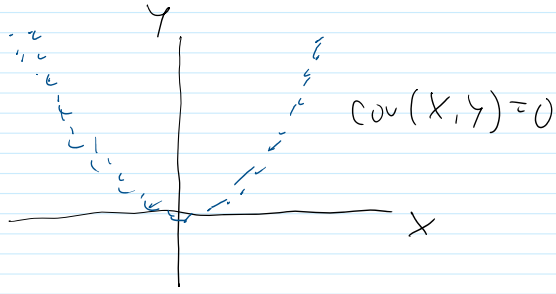
Recall that if $Cov(X,Y) = 0$ and X, Y are jointly Normal Then X, Y are independent.

- Joint Normal $\begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix} \Rightarrow$ each X_i is Univariate Normal
- Each X_i Univariate Normal $\not\Rightarrow \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$ Jointly Normal
- $\begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$ is joint Normal means that $\sum_{i=1}^{n} c_i X_i$ Univariate Normal $\forall c_i \in \mathbb{R}$

$$Cov(X,Y) = \mathbb{E}\left[(X - \mathbb{E}X)(Y - \mathbb{E}Y) \right]$$

↳ How linearly related X, Y are





Box-Muller we needed to change the variables

$$V = \frac{1}{2}(X^2 + Y^2)$$

$$U = \frac{1}{2\pi}\arctan(Y/X) + 1/2$$ ← $[-1/2, 1/2]$

$\in [0,1]$

↳ But it doesn't really matter as we only need the derivative $\frac{dU}{dX}$, $\frac{dU}{dY}$ which remove the $1/2$

Joint Density — prod of Marginal Densities

$$f_{U,V}(u,v) = f_U(u)\, f_V(v)$$ ← U, V independent

$$F_{U,V}(u,v) = F_U(u)\, F_V(v) \quad \leftarrow U, V \text{ independent}$$

$$= \underbrace{1[u \in [0,1]]}_{=1}\, e^{-v}$$

We could Do the Prob. Int. Trans since $F^{-1}(\cdot)$ can be numerically computed in R or Python

```
# PIT using qnorm
norm.pit <- function(n){
  return(
    qnorm( runif(2*n) )
  )
}
```

u

qnorm(.) = "Quantile" Function

← Do Box-Muller but in C code

```
case BOX_MULLER:
    if(BM_norm_keep != 0.0) { /* An exact test is intentional */
        s = BM_norm_keep;
        BM_norm_keep = 0.0;
        return s;
    } else {
        theta = 2 * M_PI * unif_rand();
        R = sqrt(-2 * log(unif_rand())) + 10*DBL_MIN; /* ensure non-zero */
        BM_norm_keep = R * sin(theta);
        return R * cos(theta);
    }
```

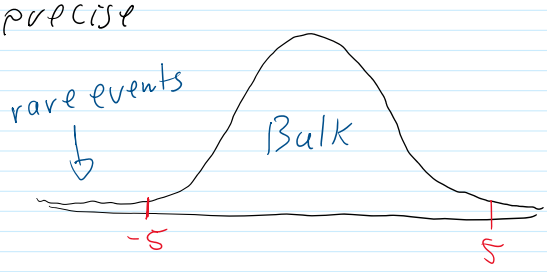
← Do PIT in C code

```
case INVERSION:
define BIG 134217728 /* 2^27 */
    /* unif_rand() alone is not of high enough precision */
    u1 = unif_rand();
    u1 = (int)(BIG*u1) + unif_rand();
    return qnorm5(u1/BIG, 0.0, 1.0, 1, 0);
```

↳ Even if the Math Formulae make sense we always have to think about the finiteness of computers.

→ When we check for goodness of fit, we usually do so in the bulk of the distribution. Dealing with very rare event / values can be hard to get precise

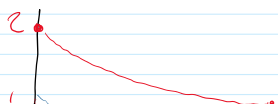
- See Extreme Value Theory
- Numerical precision is tricky!



Another Way to generate an Exp r.v.

- PIT: if $U \sim Unif[0,1]$ then $X = -\frac{1}{\lambda} \log U \sim Exp(\lambda)$ $\quad \lambda > 0$
- Truncated Exponential $X_0 \in [0, \log 2]$

$$F_{X_0}(x) = 2e^{-x}$$



$$f_{X_0}(x) = 2e^{-x}$$

$$F_{X_0}(x) = 2 - 2e^{-x}$$

* Why? working on a bounded interval is often nicer than working on $\mathbb{R}^+$

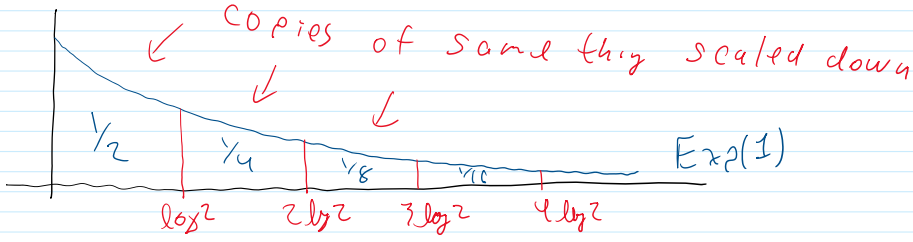
We can copy the truncated Exp to get a stand Exp

$$K \sim \text{Geometric}(1/2), \quad P(K=k) = (1/2)^{k+1}$$

For $k = 0, 1, 2, \ldots$

If $X_0 \perp\!\!\!\perp K$ then we claim that

$$X := X_0 + K \log 2 \sim \text{Exp}(1)$$



$$F_X(x) = P(K=k)\, F_{X_0}(x_0)$$

$$= \underset{\text{Geo}}{2^{-k-1}} \; \underset{\text{truncated}}{2e^{-x_0}} = \exp(\underbrace{-k\log 2 - x_0}_{-x}) = e^{-x}$$